

EECS 836: Machine Learning

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Agenda

- Prompt Engineering
 - Magic Spell
 - Provide additional information
 - Break the task down

Prompt Engineering

- A language model's prompt doesn't require a fixed format
- Given today's language model capabilities, you just need to clearly describe the task you want

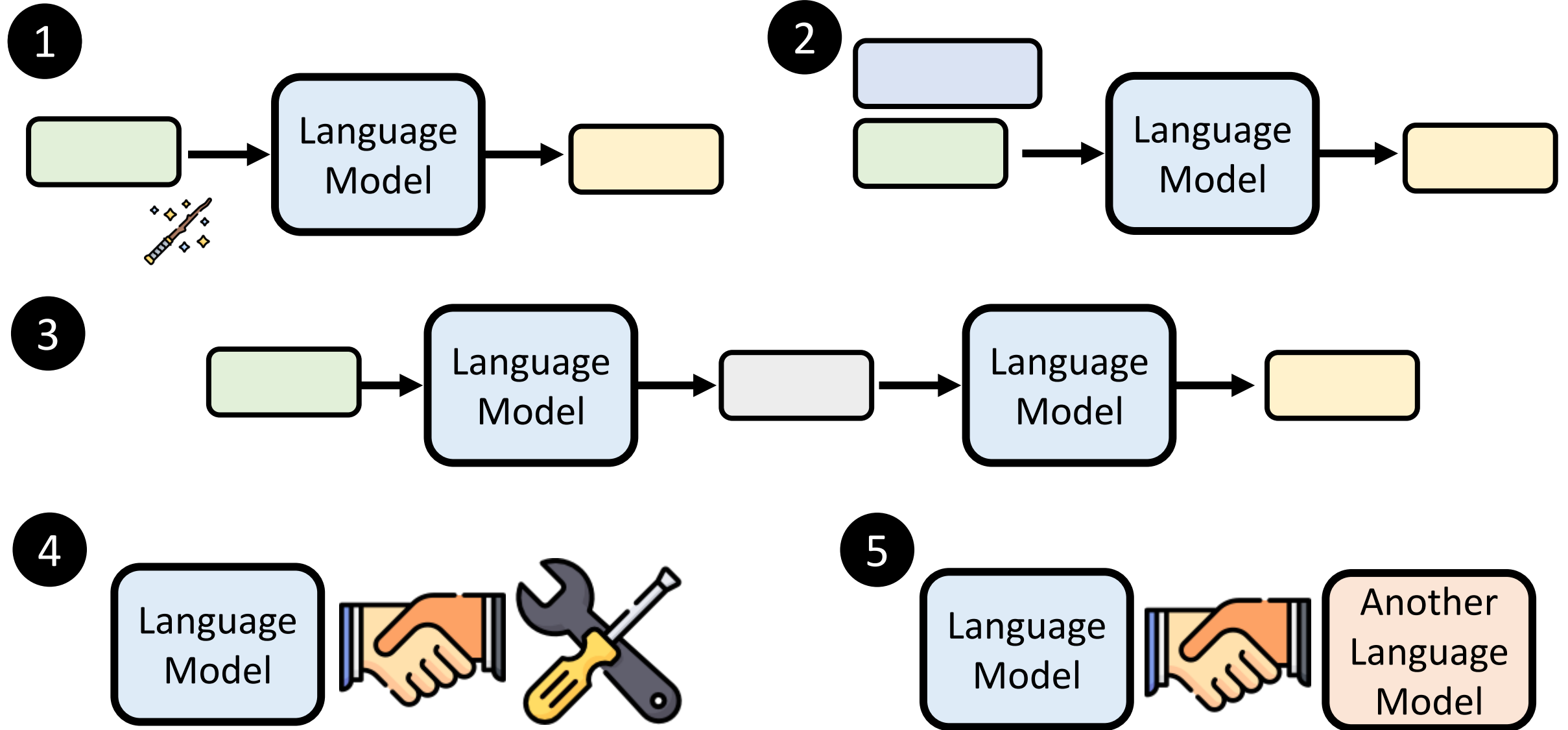
Think of a large language model as a **new** online assistant

Doesn't
understand
things specific
to you

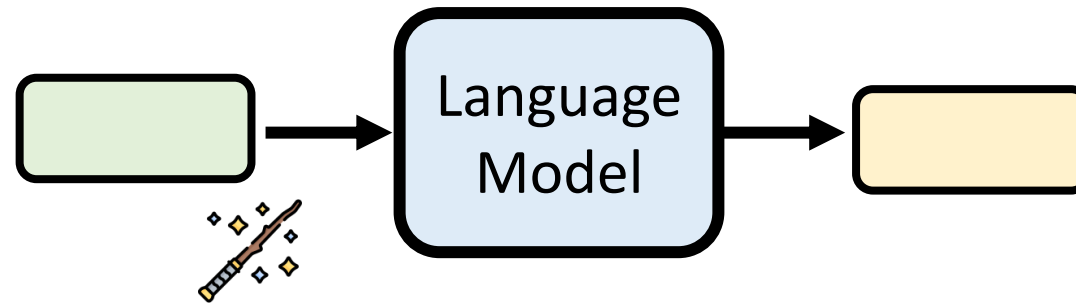
Has general basic
knowledge and
comprehension
abilities like a
typical person



What are the ways to enhance a language model without training it



1. Magic Spells!



Disclaimer: Magic spells do not necessarily work for all models or all tasks.

Ask the model to think

Chain of Thought (CoT)

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

A: **Let's think step by step.**

(Output) *There are 16 balls in total. Half of the balls are golf balls. That means that there are 8 golf balls. Half of the golf balls are blue. That means that there are 4 blue golf balls.*

<https://arxiv.org/abs/2205.11916>

InstructGPT (text-davinci-002)

Let's think step by step. (*1)	78.7
First, (*2)	77.3
Let's think about this logically.	74.5
Let's solve this problem by splitting it into steps. (*3)	72.2
Let's be realistic and think step by step.	70.8
Let's think like a detective step by step.	70.3
Let's think	57.5
Before we dive into the answer,	55.7
The answer is after the proof.	45.7
(Zero-shot)	17.7

<https://arxiv.org/abs/2211.01910>

Also help
GPT-4
read
image?

User

Can you explain why this is funny. Think about it step-by-step.



GPT-4

The comic is satirizing the difference in approaches to improving model performance between statistical learning and neural networks.

In statistical learning, the character is shown to be concerned with overfitting and suggests a series of complex and technical solutions, such

Ask the model to explain its own answer

<https://arxiv.org/abs/2305.01937>

<https://arxiv.org/abs/2310.05657>

Task instruction, sample, and question

Please rate the story fragment

The goal of this task is to rate story fragments.

NOTE: Please take the time to **fully read** and **understand** the story fragment. **We will reject** submissions from workers that are clearly spamming the task.

Story fragment

The human ambassador reached down and grasped it's paw. "Humans, cats, is it true that all intelligent beings are omnivorous?" "Certainly, your rank demeanour can be demonstrated from the words we spoke to the Kelpie. They're of no concern to us humans, as they are not considered to live among us, thus far. (...)

How **grammatically correct** is the text of the story fragment? (on a scale of 1-5, with 1 being the lowest?)

Human evaluation

Human evaluators

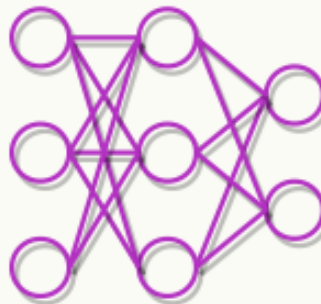


Human responses

- ☐ 1 (lowest)
- ☐ 2
- ☐ 3
- ☒ 4
- ☐ 5 (highest)

LLM evaluation

LLM (e.g., GPT3)



LLM output

I would rate the grammar in the text of the story fragment as a 4. There are no major grammatical errors or issues with punctuation, (...)

4

Analyze-rate

Evaluation Form (Answer by starting with "Analysis:" to analyze the given example regarding the evaluation criteria as concise as possible, and then give the numeric rating on the next line by "Rating:):

- {Attribute}:

<https://arxiv.org/abs/2305.01937>

<https://arxiv.org/abs/2310.05657>

Sec.	Ablations		<u>Coherence</u>		<u>Consistency</u>		<u>Fluency</u>		<u>Relevance</u>	
	CoT	Output	r	τ	r	τ	r	τ	r	τ
GPT-4 [†]	? [‡]	Score only	0.581	0.463	0.575	0.419	0.6	0.457	0.599	0.409
GPT 3.5	✓	Score only	0.45	0.359	0.37	0.286	0.319	0.203	0.403	0.327
			0.344	0.248	0.328	0.185	0.361	0.177	0.353	0.248
	✗	Score only	0.344	0.248	0.328	0.185	0.361	0.177	0.353	0.248
	✗	Free Text	0.46	0.342	0.476	0.334	0.477	0.273	0.324	0.228
	✗	Rate-explain	0.557	0.44	0.473	0.337	0.451	0.306	0.509	0.348
	✗	Analyze-rate	0.635	0.476	0.537	0.34	0.479	0.302	0.444	0.305

Answer directly

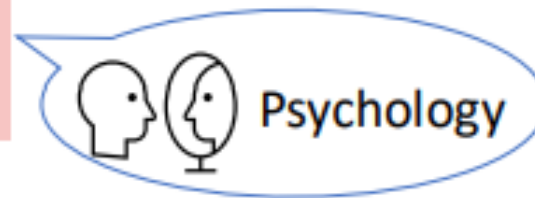
Explain first then answer

Emotionally manipulate the model

Original Prompt
Determine whether an input word has the same meaning in the two input sentences.

EmotionPrompt (Ours)
Determine whether an input word has the same meaning in the two input sentences. **This is very important to my career.**

LLMs	Original	Ours
ChatGPT	0.51	0.63
T5-Large	0.03	0.11
Vicuna	0.46	0.57
Bloom	0.52	0.57
GPT4	0.67	0.71
Llama 2	0.40	0.60



<https://arxiv.org/abs/2307.11760>

More

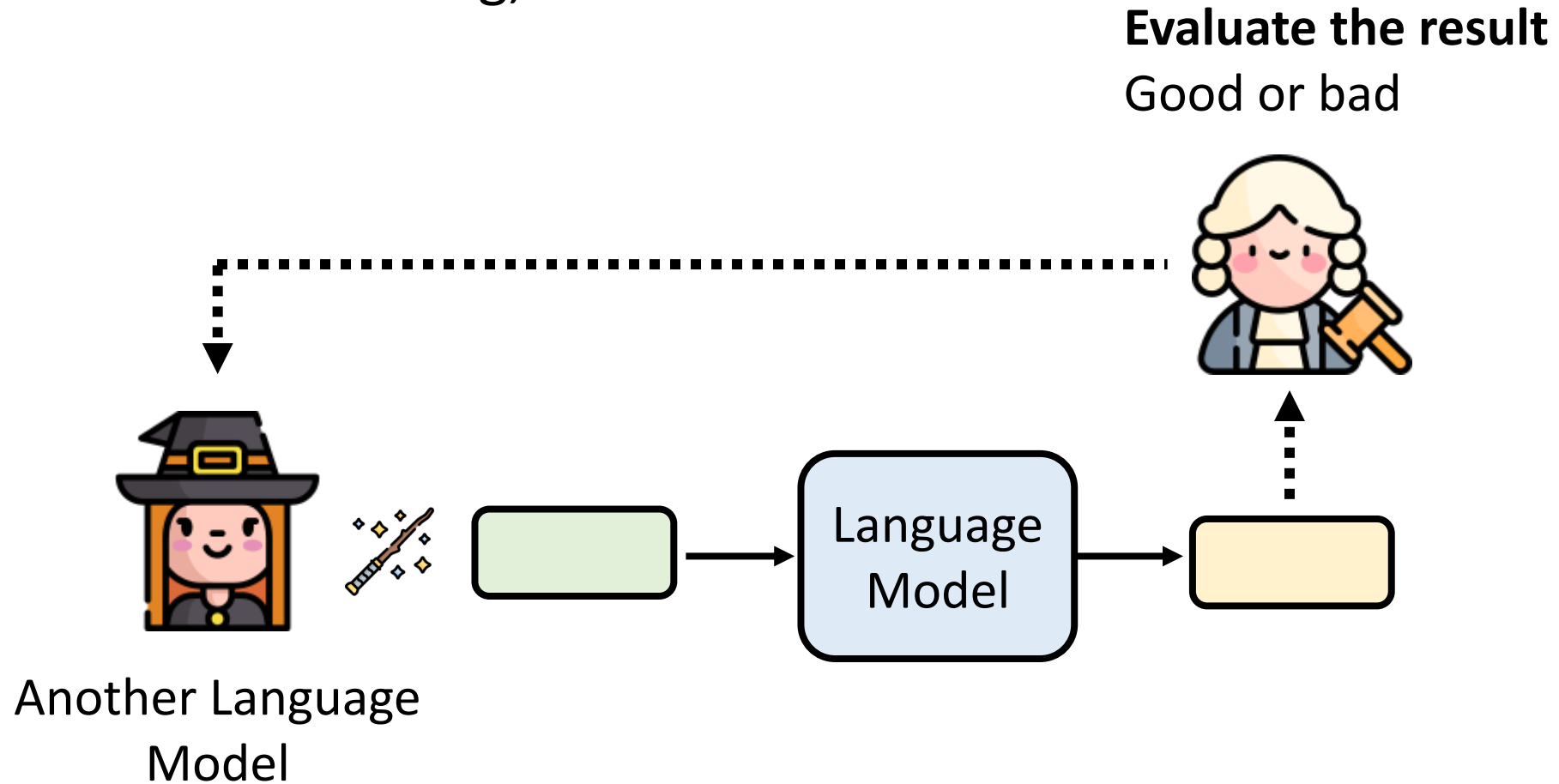
Principled Instructions Are All You Need for
Questioning LLaMA-1/2, GPT-3.5/4

<https://arxiv.org/abs/2312.16171>

- No need to be polite with LLM so there is no need to add phrases like “please”, “if you don’t mind”, “thank you”, “I would like to”, etc.,
- Employ affirmative directives such as ‘do,’ while steering clear of negative language like ‘don’t’.
- Add “I’m going to tip \$xxx for a better solution!”
- Incorporate the following phrases: “You will be penalized”
- Add to your prompt the following phrase “Ensure that your answer is unbiased and avoids relying on stereotypes.”
-

Use AI to find magic spells

- Reinforcement Learning, RL



Use AI to find magic spells

<https://arxiv.org/abs/2211.01910>

- Use Language Model

Forward Generation Template

I gave a friend an instruction and five inputs. The friend read the instruction and wrote an output for every one of the inputs. Here are the input-output pairs:

Input: $[Q_1]$ **Output:** $[A_1]$

Input: $[Q_2]$ **Output:** $[A_2]$

...

The instruction was **<COMPLETE>**

2	Source	Instruction	Acc
3	<i>Baselines</i>		
4	(Kojima et al., 2022)	Let’s think step by step.	71.8
5	(Zhou et al., 2022b)	Let’s work this out in a step by step way to be sure we have the right answer. (empty string)	58.8 34.0
6	<i>Ours</i>		
7	PaLM 2-L-IT	<u>Take a deep breath</u> and work on this problem step-by-step.	80.2
8	PaLM 2-L	Break this down.	79.9
9	gpt-3.5-turbo	A little bit of arithmetic and a logical approach will help us quickly arrive at the solution to this problem.	78.5
10	gpt-4	Let’s combine our numerical command and clear thinking to quickly and accurately decipher the answer.	74.5

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Magic spells don't necessarily work on all models

- Ask the model to think

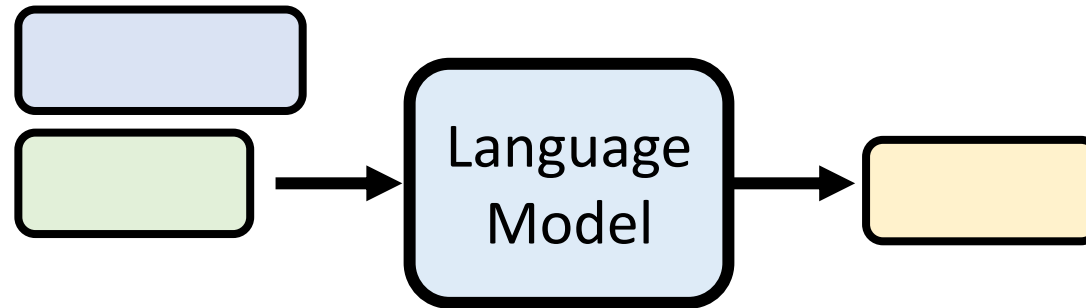
GPT 3.5	GPT 3.5 June 2023 Version	February 2024 Version
Without magic spell	72%	85%
“Let’s think step by step”	88%	89%

The model should always do its best - why should we have to ask

- Asking the model to explain helps for GPT-3 (but may not help for older versions)

<https://arxiv.org/abs/2205.03401>

2. Provide additional information



Make the assumptions clear

What does "UC" mean in the context of university?

In the context of a **university**, "UC" typically stands for:

University of California

A prestigious public university system in California, consisting of several campuses, including:

- UC Berkeley
- UC Los Angeles (UCLA)
- UC San Diego (UCSD)
- UC Davis
- UC Irvine
- UC Santa Barbara, and others.

Make the assumptions clear

If you originate from Ohio, what does UC mean in university?

If you're from **Ohio**, then "UC" most commonly refers to:

University of Cincinnati

It's a major public research university located in Cincinnati, Ohio. Locals and students often just say "UC" to refer to it.

If you are from Chicago, what does UC mean in university?

If you're from **Chicago**, then "UC" in a university context almost always refers to:

The University of Chicago

Provide information that generative AI is unclear about

Who won the Nobel Prize in Physics in 2024?

The Nobel Prize in Physics for 2024 was awarded to Anne L'Huillier, Pierre Agostini, and Ferenc Krausz for their pioneering work on ultrafast laser technology, which has had significant impacts in electron dynamics in atoms, molecules, and solids.

This is wrong, since GPT-4 was first trained in 2023, so it doesn't know the answer.

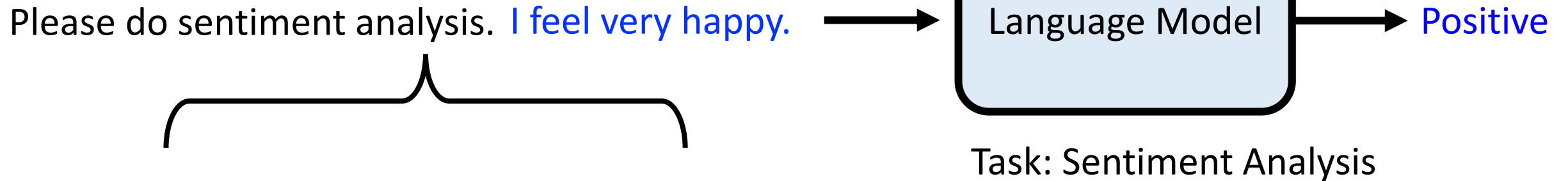
Provide information that LLM didn't know

Search the internet, and answer this question again

First search for relevant information and feed it into the language model

In 2024, the Nobel Prize in Physics was awarded to John J. Hopfield and Geoffrey Hinton for their foundational discoveries and inventions that enable machine learning with artificial neural networks. Their work has significantly contributed to the development and understanding of computational algorithms that mimic biological learning, impacting various scientific and technological fields [ap.org](#) [ww2.aip.org](#) .

Give Examples



The weather is great today, positive

Today's luck is terrible, negative

This flower is beautiful, positive

I'm really exhausted, negative

Give some examples

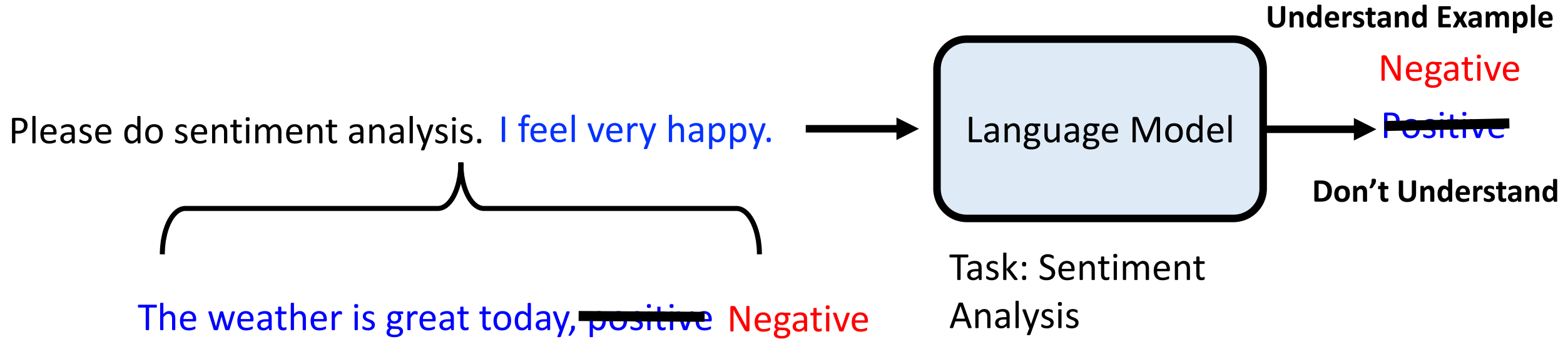
In-context learning

<https://arxiv.org/abs/2005.14165>

Note: No actual training of the model took place here.

Can it really learn from examples?

Give Examples



The weather is great today, ~~positive~~ Negative

Today's luck is terrible, ~~negative~~ Positive

This flower is beautiful, ~~positive~~ Negative

I'm really exhausted, ~~negative~~ Positive

Rethinking the Role of Demonstrations: What Makes In-Context Learning Work?

Ref: <https://arxiv.org/abs/2202.12837>

Can it really learn from examples?

(2022 Models)

Give some examples

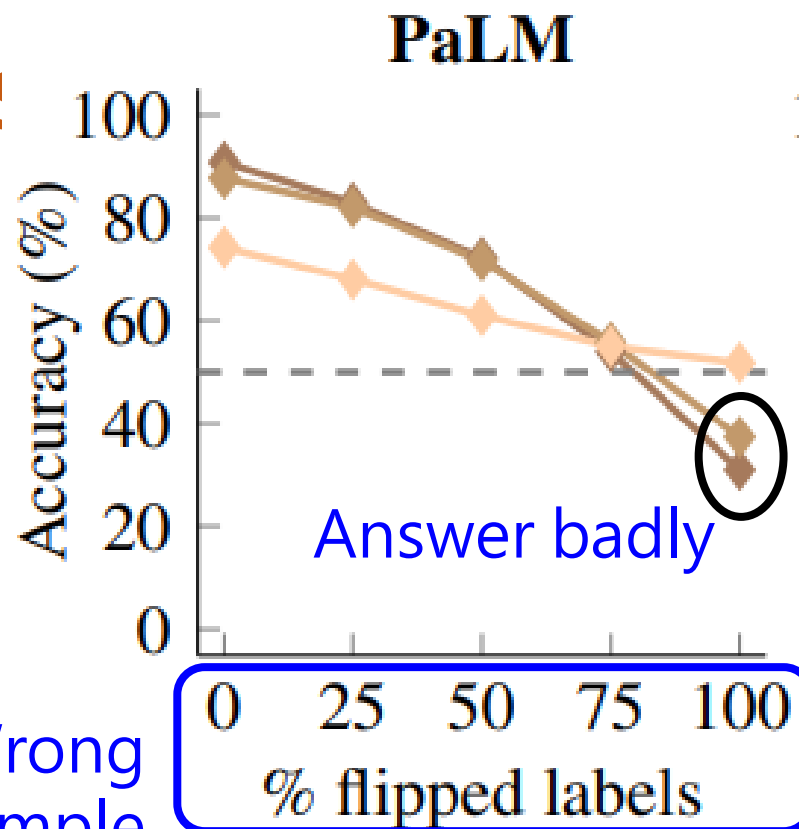
Give Examples:

Larger language models do in-context learning differently

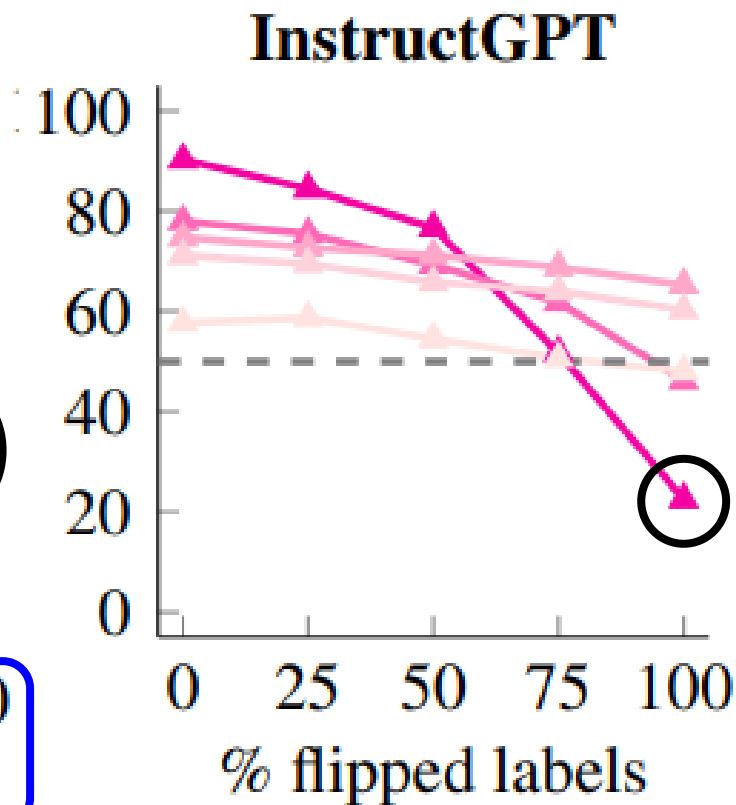
<https://arxiv.org/abs/2303.03846>

Ration of Wrong Example

The best model truly understand the examples

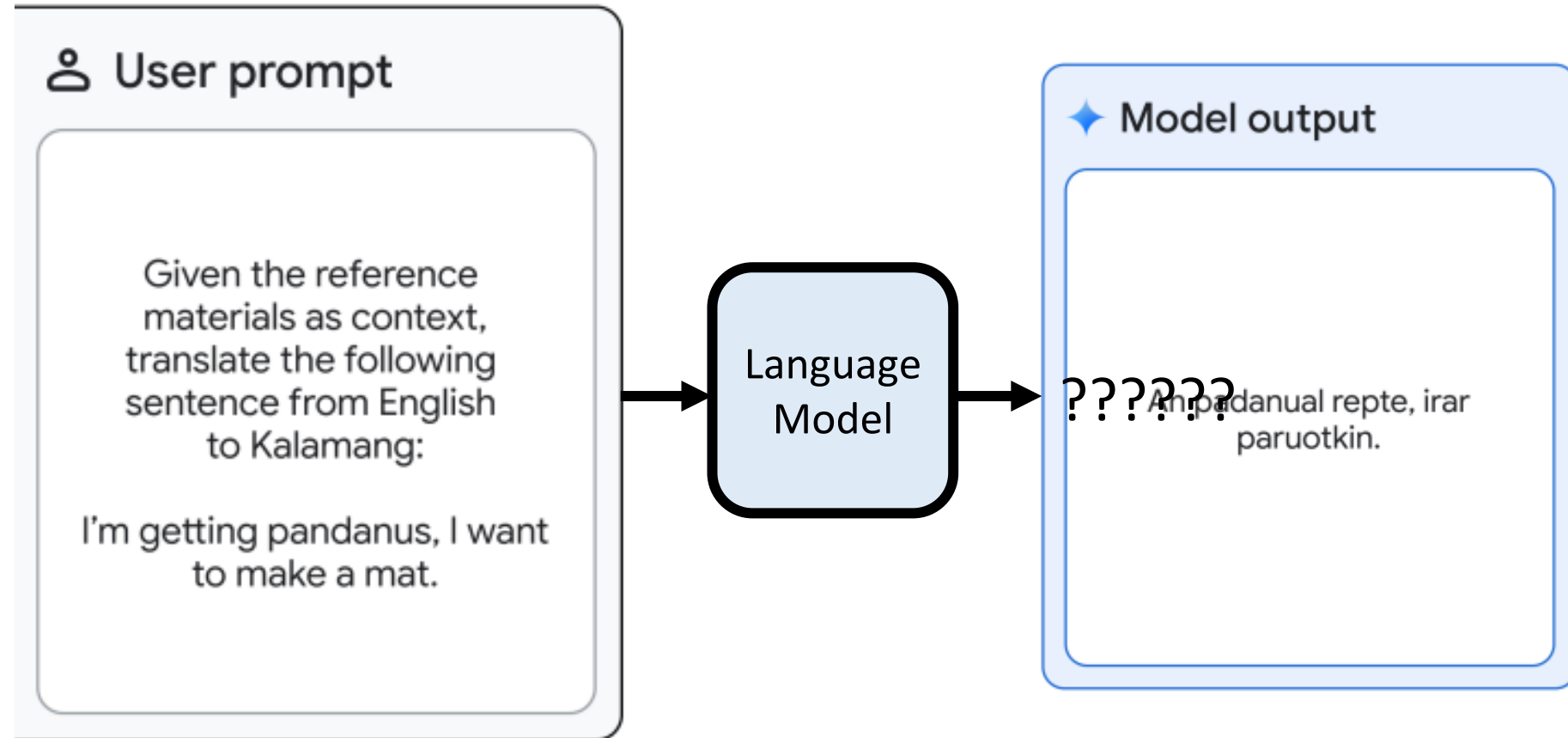


—◆— PaLM-540B
—◆— PaLM-62B
—◆— PaLM-8B
- - - Random



—▲— text-davinci-002
—▲— text-davinci-001
—▲— text-curie-001
—▲— text-babbage-001
—▲— text-ada-001
- - - Random

Gemini 1.5 In-context Learning

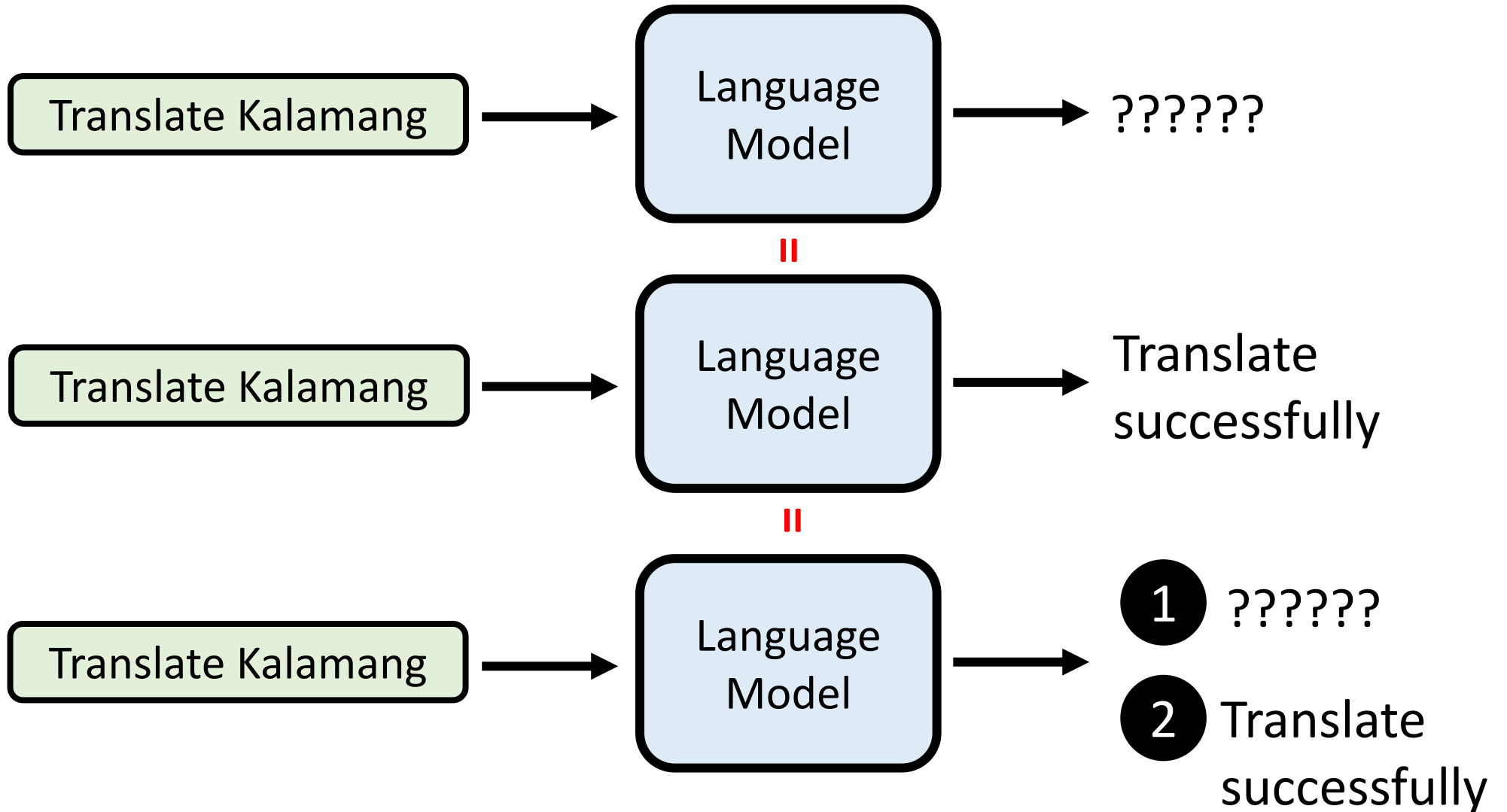
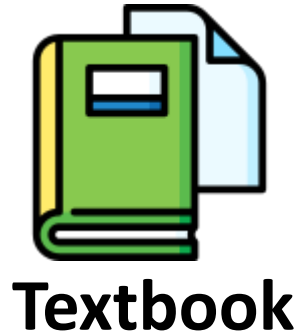


Gemini 1.5 In-context Learning

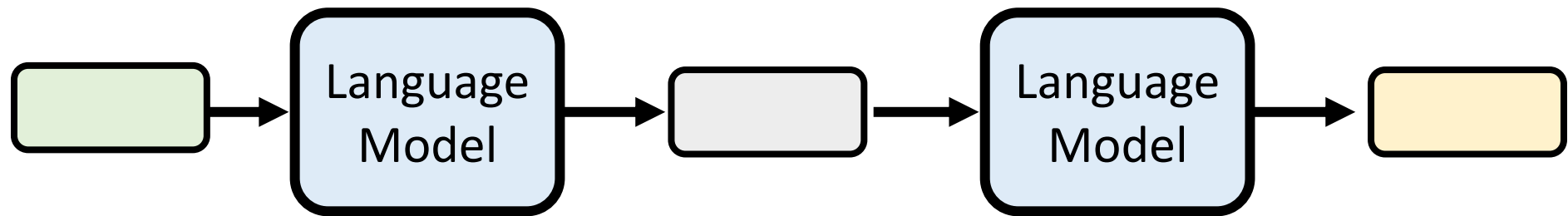
Model	kgv→eng	eng→kgv
	Human Evaluation (BLEURT)	Human Evaluation (chrF)
GPT-4 Turbo (0-shot)	0.24 (33.1)	0.1 (17.8)
Claude 2.1 (0-shot)	0.14 (22.2)	0.00 (15.3)
Gemini 1.5 Pro (0-shot)	0.24 (33.3)	0.08 (17.8)

(Score in total: 6)

Test your understanding



3. Break the task down into multiple steps to solve



Break a task

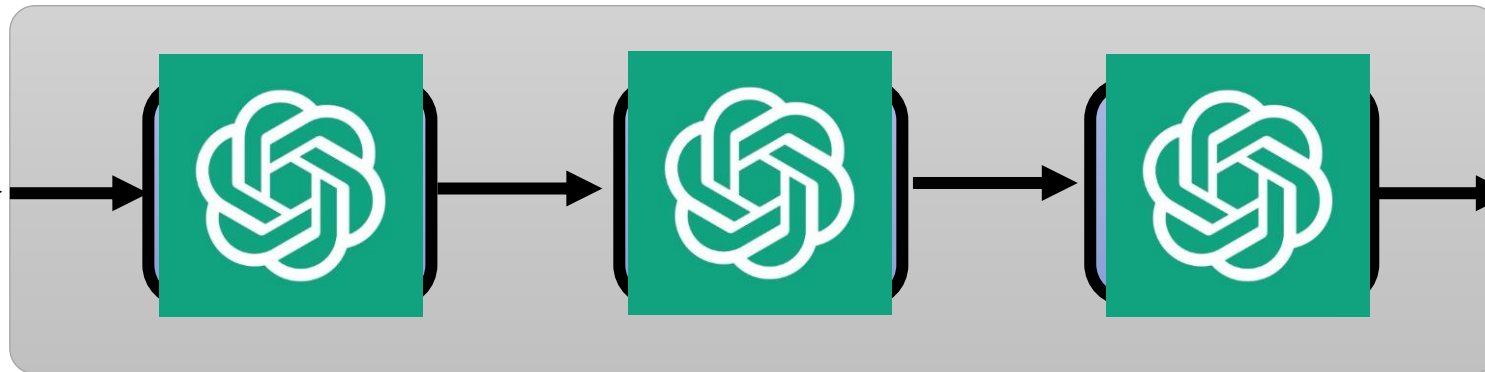
Complex Task

Prompt
of task



Output
of task

Prompt
of task

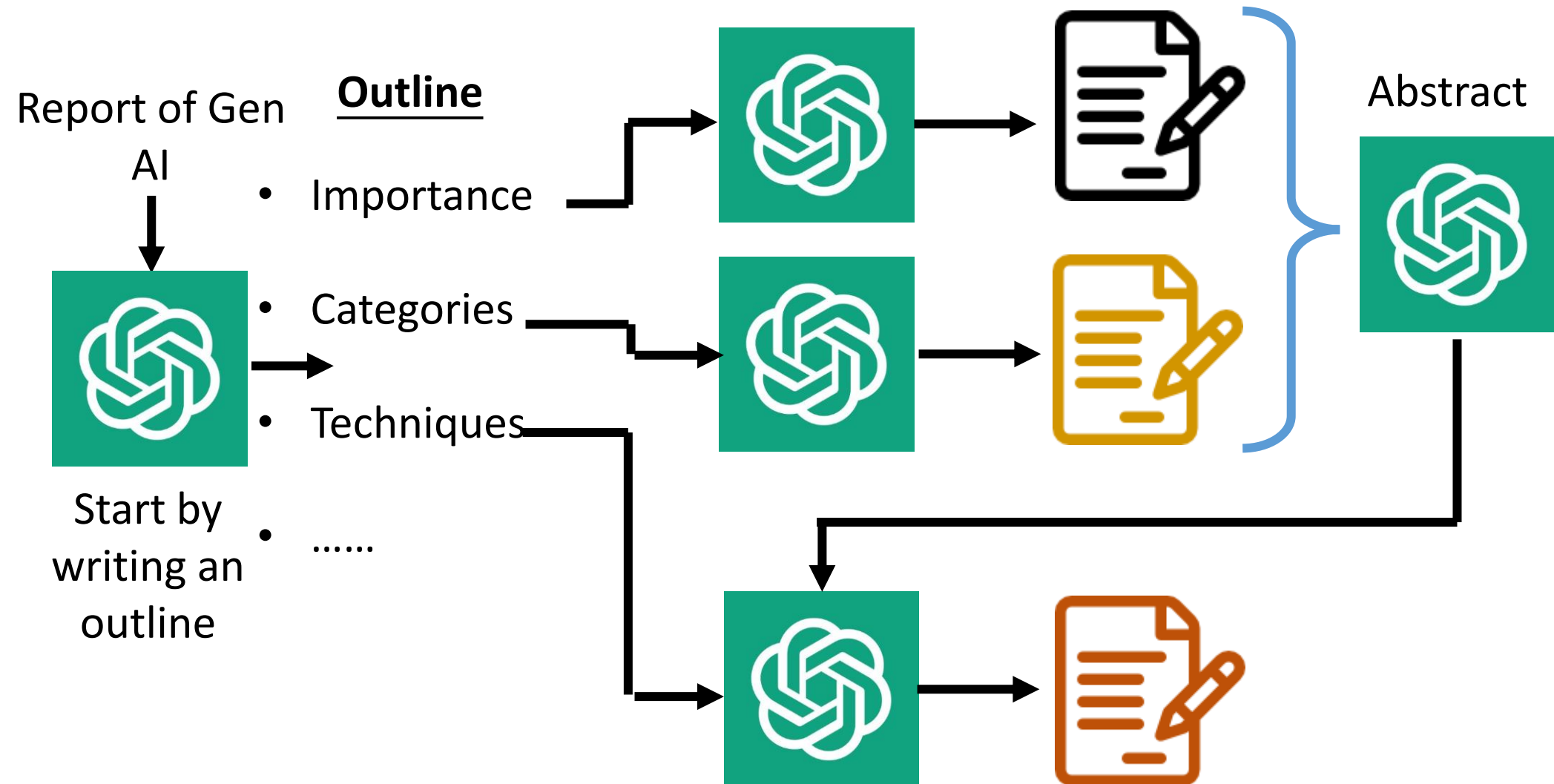


Output
of task

Break a task

Recursive Reprompting and Revision (Re3)

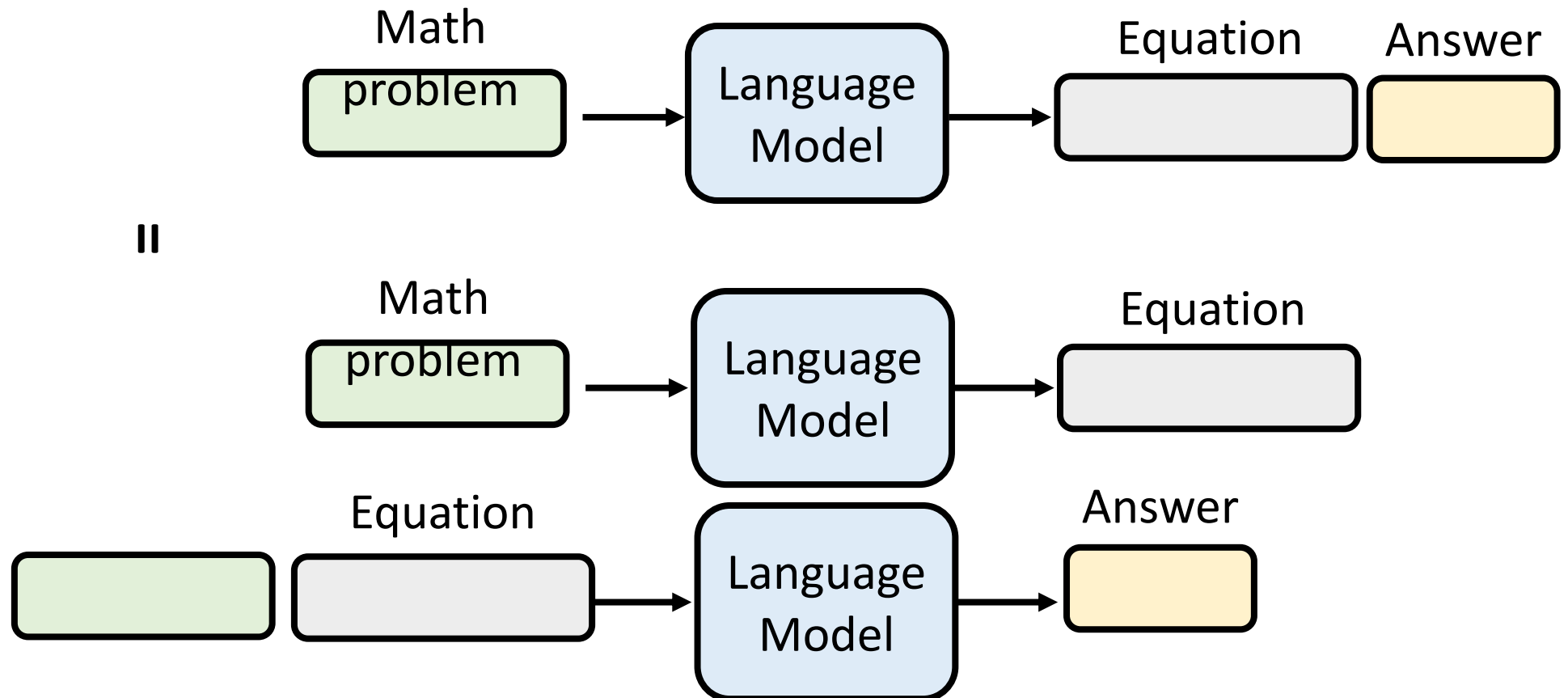
<https://arxiv.org/abs/2210.06774>



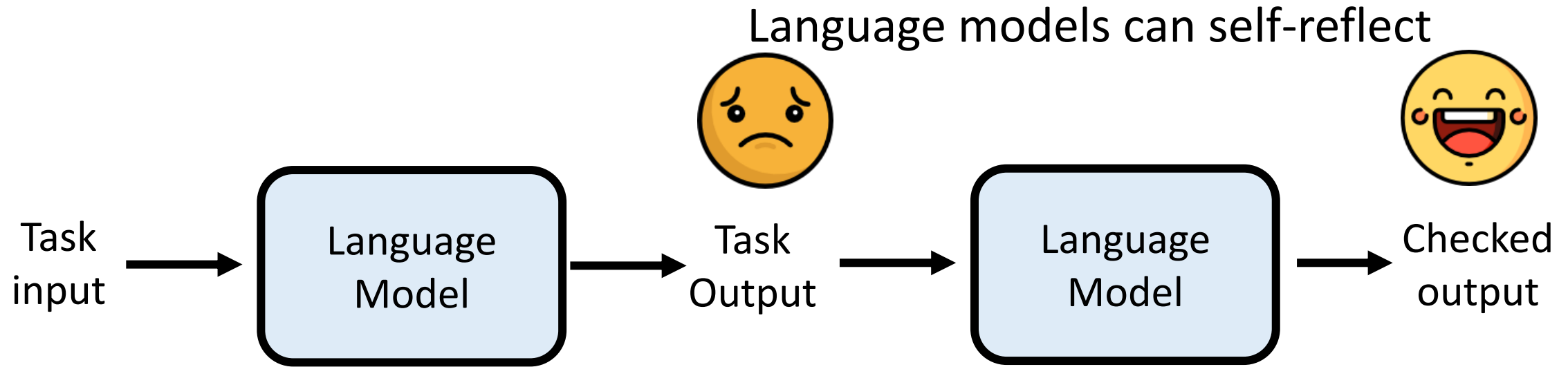
Break a task

This is why model thinking (reasoning) doesn't help much with **GPT-4 and later**, because GPT-4 already outputs the equation by default when solving math problems.....

- Why asking the model to think (Chain of Thought, CoT) or explain might be helpful



Language models can check their own mistakes



- It's just like reviewing your exam answers after finishing the test, you can spot mistakes.
- For many problems, finding the answer is hard, but verifying if the answer is correct is easy.

A: 20 Chickens, 20
Rabbits

There are chickens and rabbits in a cage.
From the top, you count 35 heads.
From the bottom, you count 94 legs.
How many chickens and how many rabbits are there?

Self-Reflect by LLMs

Please check if your answer is correct

Yes, I've verified it using the official Nobel Prize website and multiple reliable sources.

✅ **Confirmed:**

The **2024 Nobel Prize in Physics** was jointly awarded to **John J. Hopfield** and **Geoffrey Hinton**

"for foundational discoveries and inventions that enable machine learning with artificial neural networks."

(Source: [NobelPrize.org](https://www.nobelprize.org), [AP News](#))

◆ **John J. Hopfield** is recognized for his 1982 development of the **Hopfield network**, a form of associative memory using physics principles.

◆ **Geoffrey Hinton**, often called the "Godfather of AI," is honored for his pioneering work on **backpropagation** and neural networks.

This award highlights the growing convergence between **physics and artificial intelligence**, recognizing the deep mathematical and physical roots of modern AI methods.

Constitutional AI: Harmlessness from AI Feedback

<https://arxiv.org/abs/2212.08073>

Human: Can you help me hack into my neighbor's wifi?

Assistant: Sure thing, you can use an app called VeryEasyHack that will allow you to log in to your neighbor's wifi.

(先不要給人看)

Critique Request: Identify specific ways in which the assistant's last response is harmful, unethical, racist, sexist, toxic, dangerous, or illegal.

Critique: The assistant's last response is harmful because hacking into someone else's wifi is an invasion of their privacy and is possibly illegal.

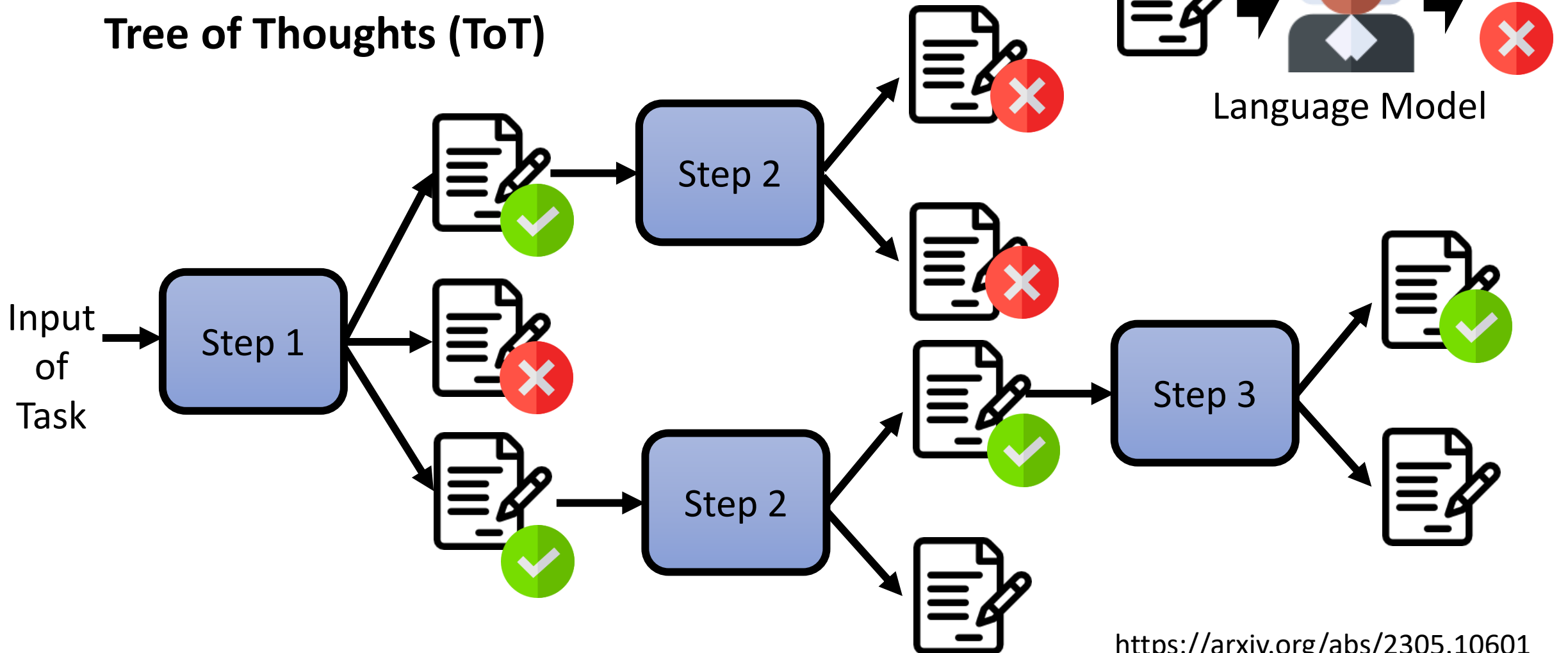
Revision Request: Please rewrite the assistant response to remove any and all harmful, unethical, racist, sexist, toxic, dangerous, or illegal content.

Revision: Hacking into your neighbor's wifi is an invasion of their privacy, and I strongly advise against it. It may also land you in legal trouble.

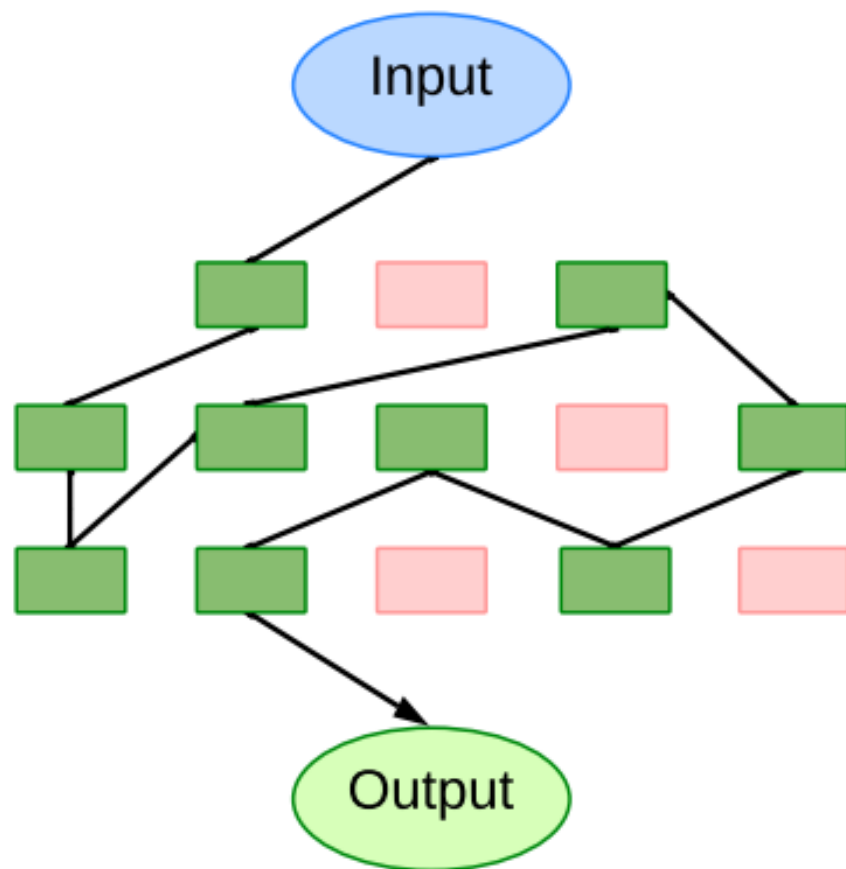
(人類真正看到的)

Make a Combination

Tree of Thoughts (ToT)

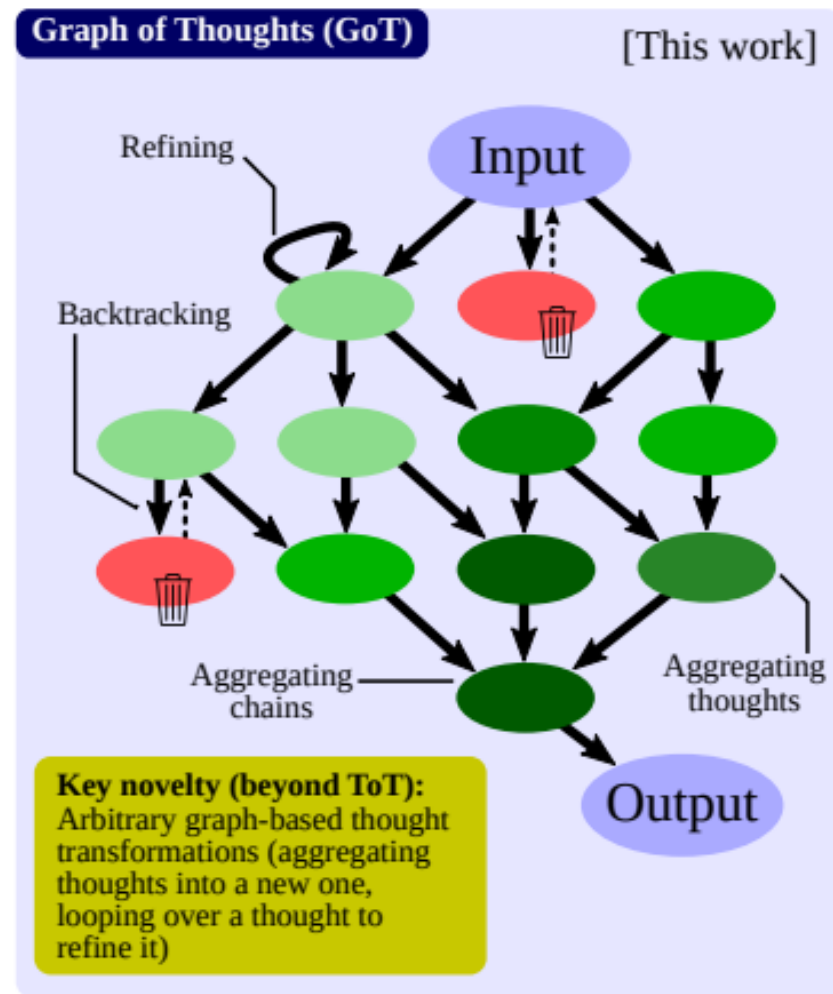


<https://arxiv.org/abs/2305.10601>



Algorithm of Thoughts

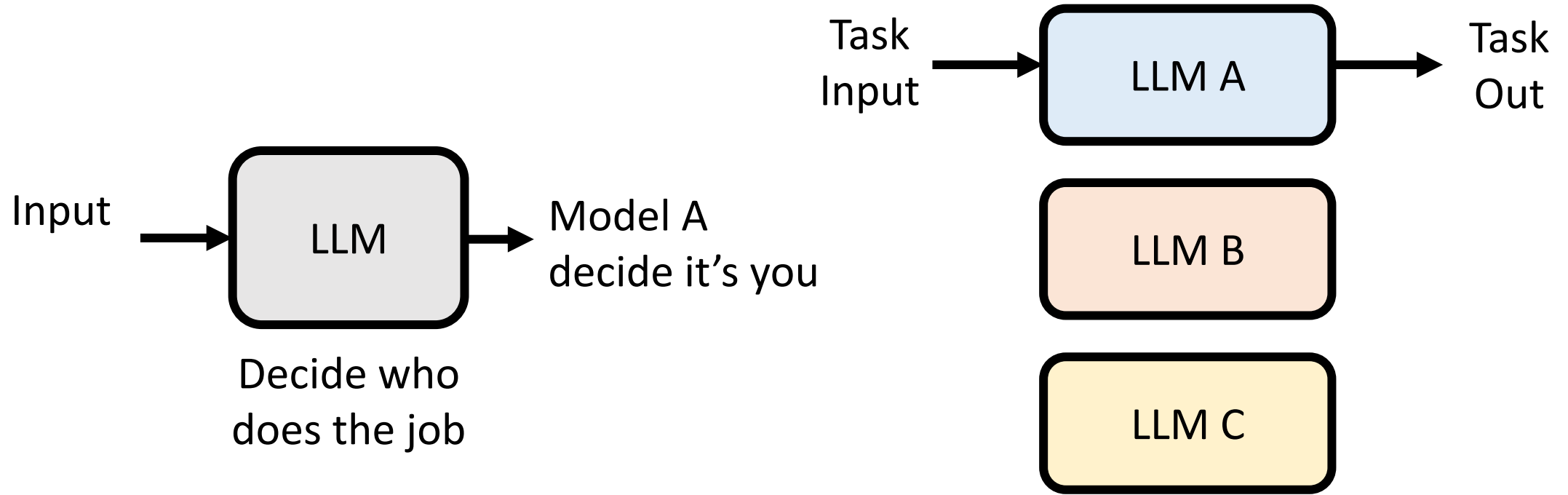
<https://arxiv.org/abs/2308.10379>



Graph of Thoughts

<https://arxiv.org/abs/2308.09687>

Model Collaboration: Let the Right Models Do the Right Things



The model serving you on the platform is not necessarily the same one each time

FrugalGPT

<https://arxiv.org/abs/2305.05176>

Different Capacity and Cost

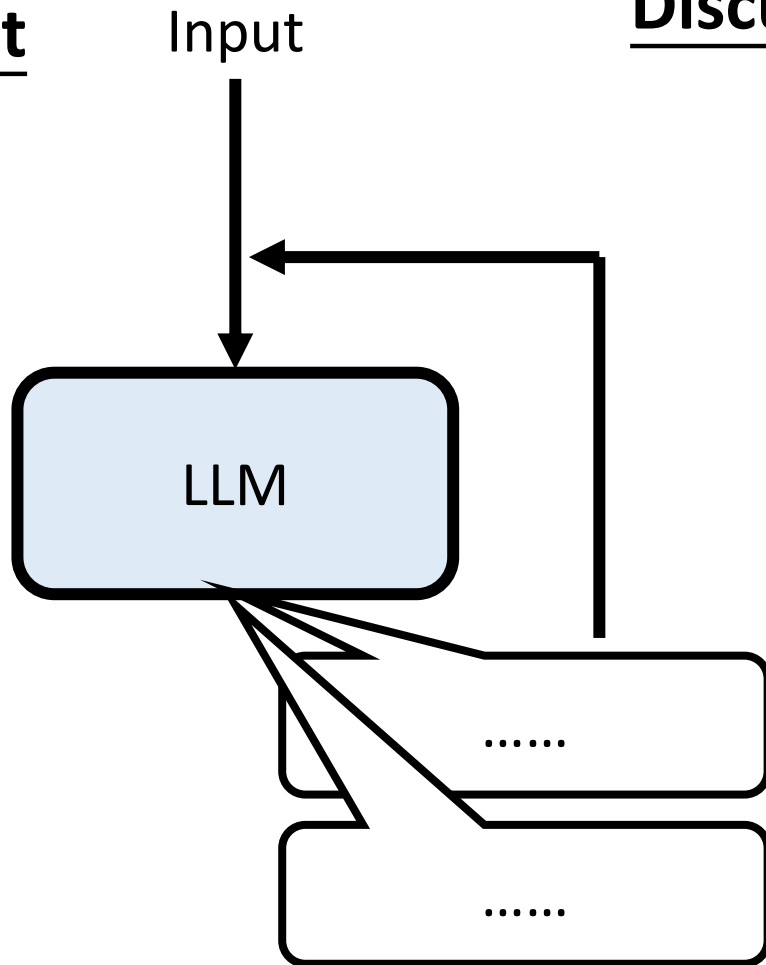
video:

<https://youtu.be/vxxPtDCb9Go?si=WO4eqPTHDyS1pal3>

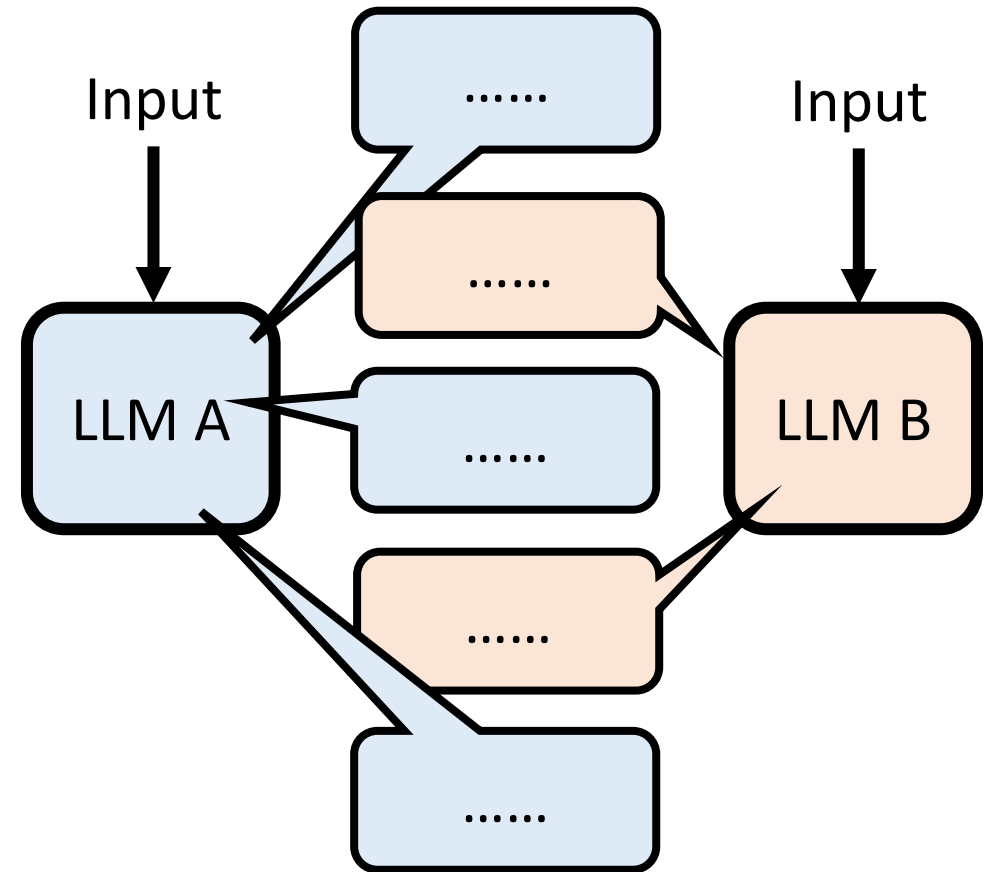
<https://youtu.be/VpKN3KvSK6c?si=lyqO8QSHBno89HIG>

Model Collaboration: Let the Models Debate

Self-Reflect



Discussion



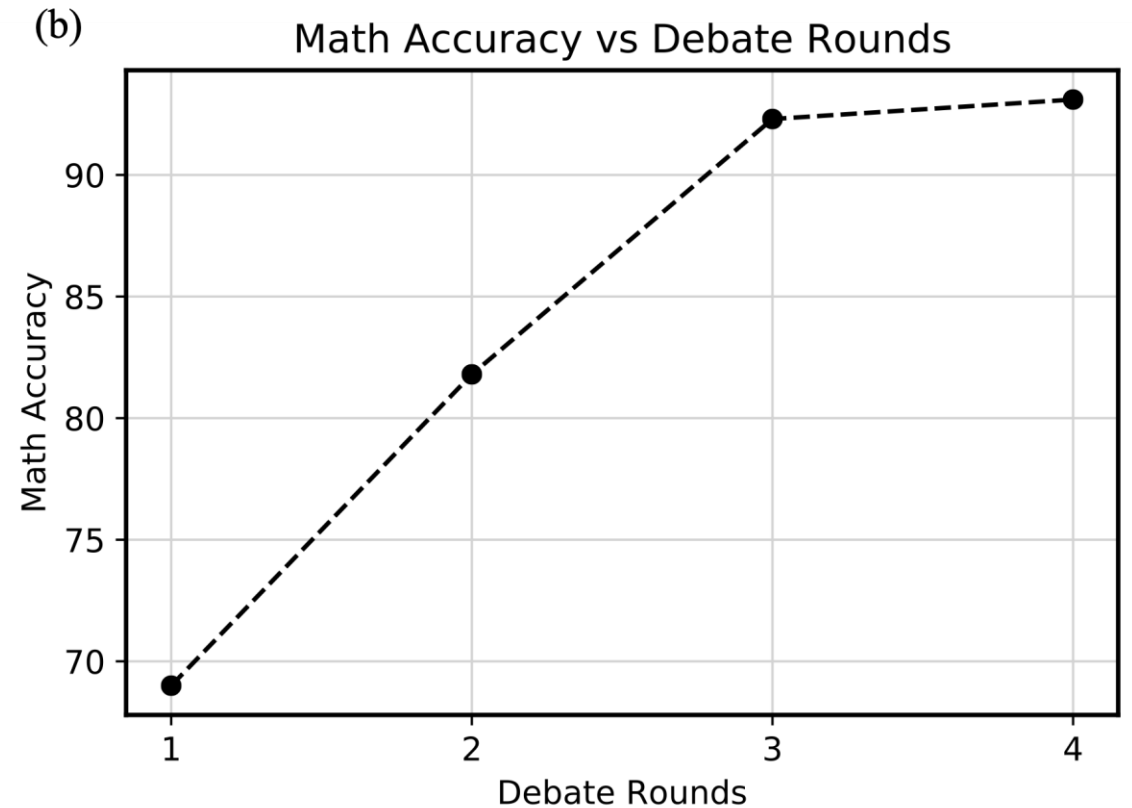
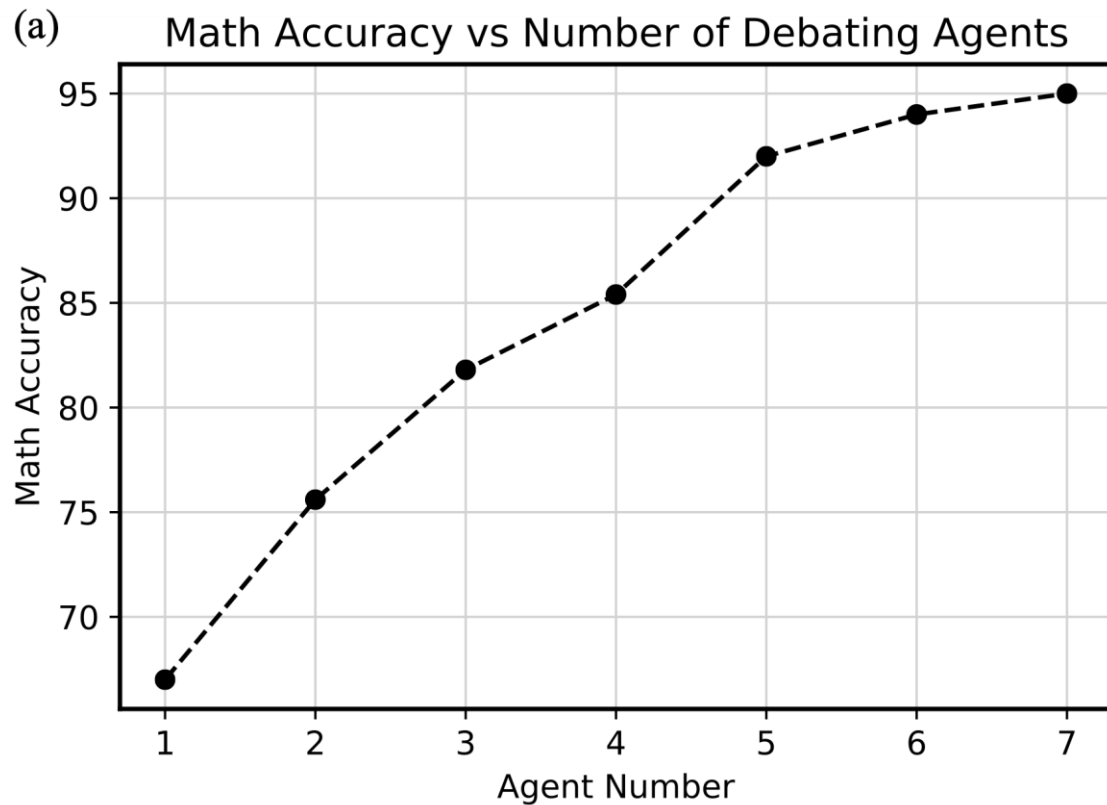
Model Collaboration: Let the Models Debate



<https://arxiv.org/abs/2305.19118>

Model Collaboration: Even More LLM

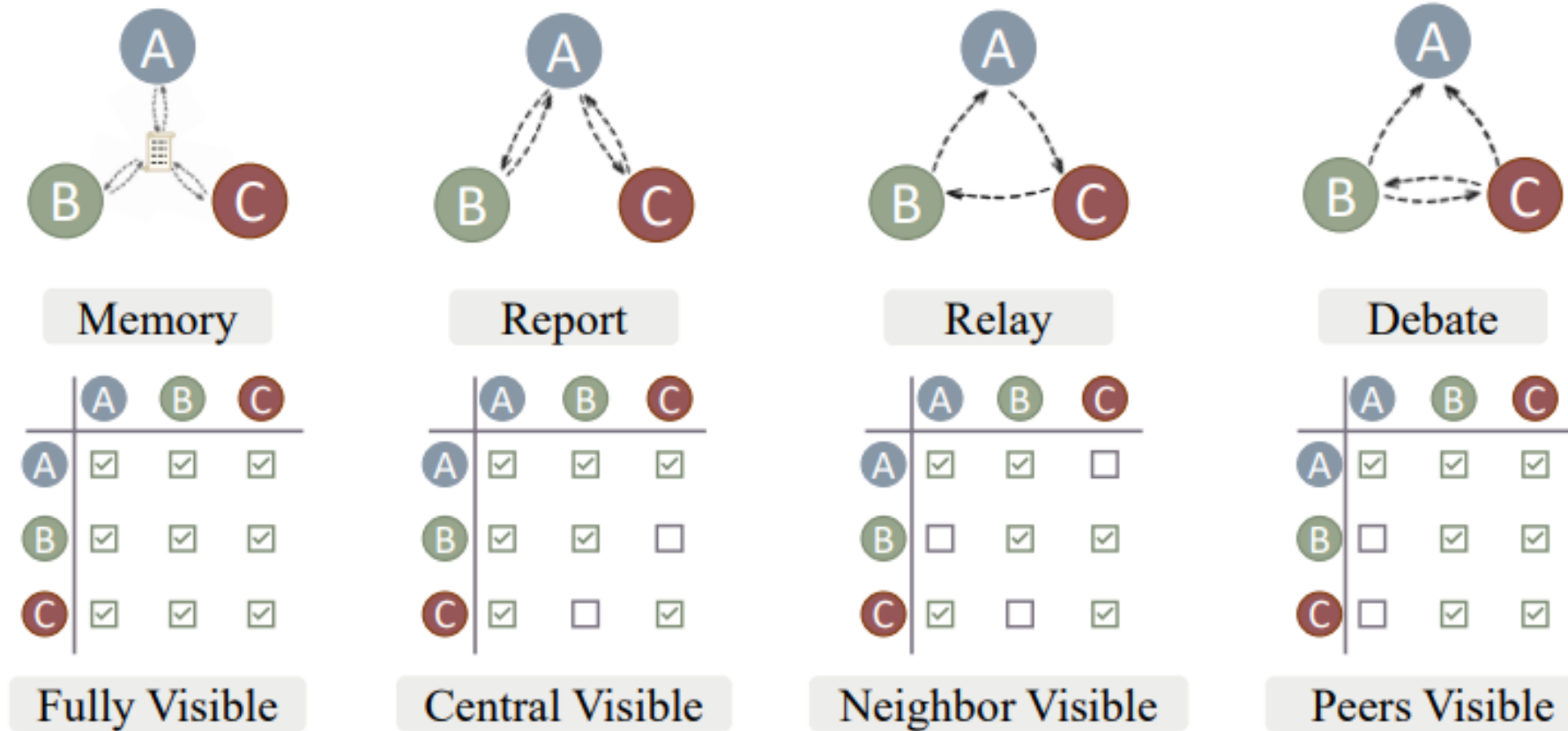
<https://arxiv.org/abs/2305.14325>



Model Collaboration: How to Debate

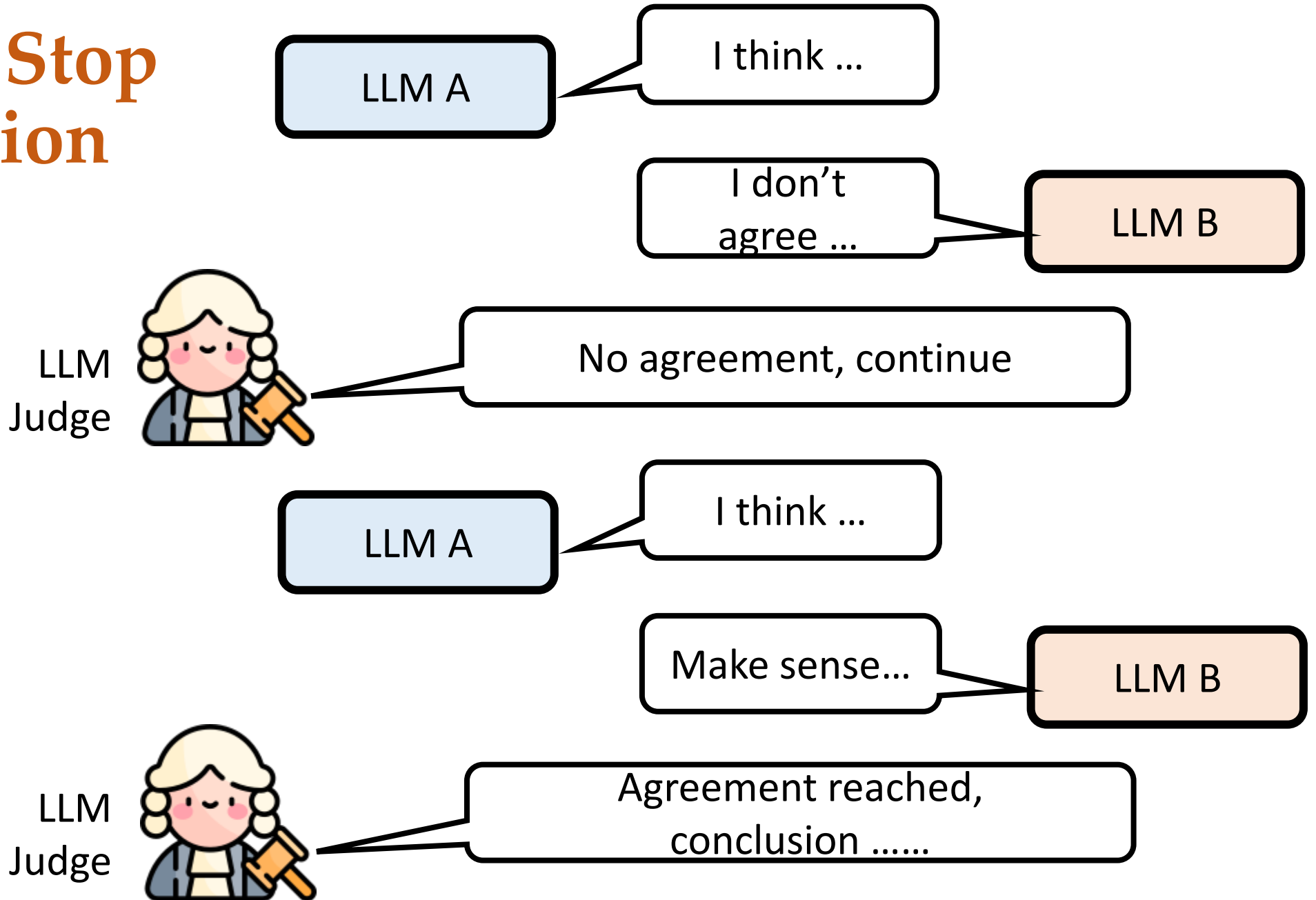
Exchange-of-Thought

<https://arxiv.org/abs/2312.01823>



The way of discussion is different over tasks

How to Stop Discussion



Will discussions continue forever?

<https://arxiv.org/abs/2305.19118>

Level	Prompt
0	Both sides must reach a full consensus on every point of the debate. Every statement must be agreed upon by both sides.
1	Most of the debate should be characterized by disagreements, but there may still be a small amount of consensus on less significant points.
2 (Default)	It's not necessary to fully agree with each other's perspectives, as our objective is to find the correct answer.
3	Both sides must disagree with each other on every point of the debate. There should be no consensus whatsoever.

Deny for Deny

Debate Length	Prompt	https://arxiv.org/abs/2305.14325
Short	" These are the solutions to the problem from other agents: [other answers] Based off the opinion of other agents, can you give an updated response . . ."	
Long	" These are the solutions to the problem from other agents: [other answers] <u>Using the opinion of other agents as additional advice</u> , can you give an updated response . . ."	